Mayank Kapadi

Paderborn, Germany | mayankkapadi97@gmail.com | +49 151 29732418
linkedin.com/in/mayank-kapadi | github.com/mayankkapadi

Professional Summary

Data and cloud-oriented engineer with hands-on experience building **hybrid physics–ML models**, scalable **data processing pipelines**, and time-series prediction systems. Strong background in Python-based data analysis, scientific computing, and ML model development, with experience combining **ODEs and neural networks** for dynamic system modeling. Familiar with cloud fundamentals and modern software practices, supported by **AWS Cloud Practitioner certification**. Proven ability to translate research concepts into **reproducible, production-ready workflows**.

Experience

Student Research Assistant, University of Paderborn, Paderborn, Germany — Apr 2025 – Present

- Designed and evaluated **hybrid ODE–ANN models** for inverse dynamics and data-driven system identification tasks.
- Developed **Python-based data pipelines** for preprocessing, training, and evaluating multivariate time-series datasets.
- Implemented physics-informed ML models using PyTorch, NumPy, and SciPy with emphasis on **reproducibility and experiment tracking**.
- Analyzed large simulation datasets and visualized results using Matplotlib to support research decisions.
- Collaborated with researchers to validate model performance and compare results against analytical and baseline approaches.

Student Research Assistant, University of Paderborn, Paderborn, Germany — Mar 2022 – Apr 2024

- Contributed to the development of an educational **Unity-based digital learning platform** with structured data handling.
- Designed reusable UI components and interaction logic using C#, Unity3D, and Git.
- Integrated backend services using HTML, PHP, CSS, and SQL for storing and retrieving application data.
- Worked in a collaborative environment to translate requirements into maintainable software modules.

Software Developer, Ashwamedh Infotech Pvt. Ltd., Mumbai, India — Jun 2018 – Dec 2020

- Developed and maintained **data-driven web applications** for enterprise payroll and management systems.
- Designed and integrated **SQL-based databases** to ensure reliable data storage and retrieval.
- Diagnosed and resolved client-side and data consistency issues, improving system reliability.
- Performed systematic testing and validation to ensure robustness of newly developed features.
- Delivered customized software solutions using HTML, PHP, CSS, SQL, and JavaScript.

Education

M.Sc. Computer Science, University of Paderborn, Germany — Apr 2021 – Sep 2025

B.E. Computer Engineering, University of Mumbai, India — Jun 2014 – May 2018

Skills

Programming & Data: Python, Pandas, NumPy, SciPy, SQL, PyTorch, C#
Cloud & Dev Foundations: AWS (EC2, S3, IAM - fundamentals), Git
Web & Scripting: HTML, CSS, JavaScript, PHP
Concepts: Data Pipelines, ETL Workflows, Cloud Fundamentals (IaaS, PaaS, SaaS), Data Modeling, Machine Learning, Time-Series Analysis, Numerical Methods

Languages

English (C1 – Business fluency)
German (B1 – Currently learning)

Projects

- **Software Architecture Concept for Vehicles (Master's Thesis, Fraunhofer IEM):** Developed a data-centric software architecture for software-intensive vehicles integrating sensors, V2X communication, security mechanisms, and AI/ML components. Created complete **SysML models in Cameo** and implemented a **MATLAB/Simulink simulation** to validate system behavior and data flow.
- **Adaptive Virtual Chemistry Laboratory (VIRTUCHEMLAB):** Implemented a collaborative VR-based digital twin with structured data handling for simulations, including interaction logic, UI components, and 3D environments using Unity3D, Blender, C#, Python, and Git.
- **Smart Grid Technology – Monitoring & Prediction (B.E. Project):** Designed a data-driven smart grid system to monitor household electricity consumption and predict future usage using algorithmic forecasting methods.

Certifications

AWS Certified Cloud Practitioner (Credly)
Microsoft MTA Security Fundamentals (Credly)